4 Installation

4.1 Installation Site

Illustration 12 shows the installation diagram of the machine and its associated equipment. Dimension are in Millimeters.

An additional space of about 1 m around the grinding installation has to be allowed for maintenance and service operations.

Space requirement:

Machine and system components: approx. 54 m^2;

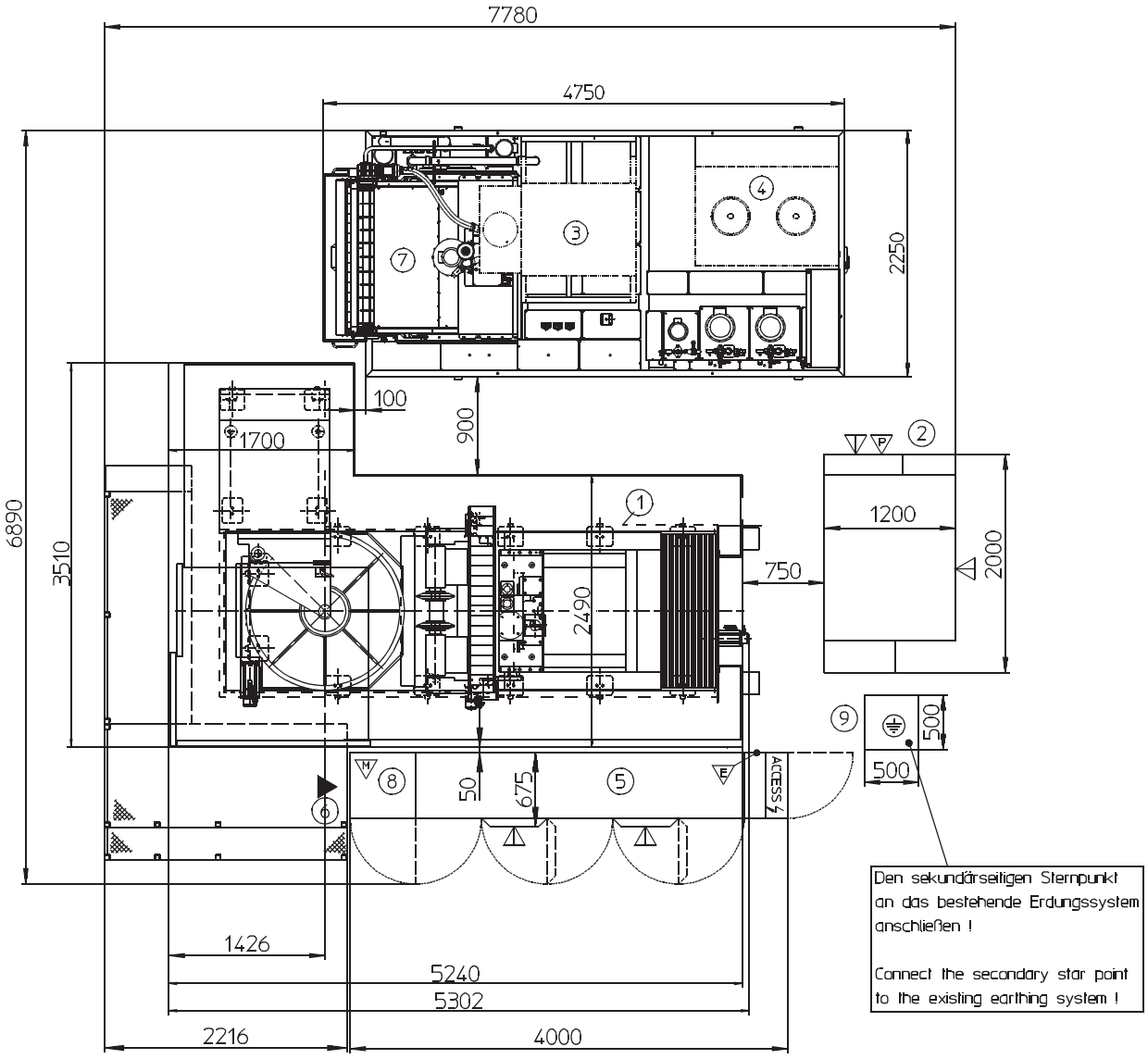
with additional space: approx. 87 m^2;

Care has to be taken that the recooler of the coolant filtration system can intake sufficient cool air. This necessitates

- the provision of adequate room between wall and adjacent large machinery;
- that the installation site is not too close to sources of heat causing the recooler to take in air that is too warm;
- that the air taken in is reasonably clean to avoid premature loading of the recooler's air filter mat and loss of cooling power.

Foundation load specification

NILES recommends installing the machine on a foundation with minimum dimensions given in the foundation diagram per Illustration 13, thickness of the concrete about 600 mm, and extending beyond the line of the support plates about 600 mm.
When installing the machine in a pit, then its base area as well as the side walls are to be sealed with oil-tight resin upto 100 mm in height.



Legend:

1: Grinding machine
2: Hydraulic package
3: Oil-mist separator
4: Coolant unit
5: Electrical cabinet
6: Control panel
7: Grinding oil filter
8: Printer
9: Transformer

Dims in mm

Illustration 12: Installation diagram

E: Electrical connection of the machine (as for specification, see chapter 4.3.1)
P: Pneumatic connection (As for specification, see chapter 4.3.2)
M: Modem connection
⏚ Grounding

4.2 Load specification of the foundation

The machine weight is approximately 25 500 kg to be considered for the foundation load specification.

The minimum thickness of the concrete plate is recommended to amount to 600 mm.The concrete plate dimensions are shown in Illustration 13.

The concrete is to be prepared according to the concrete group BI, strength category B25 (rated strength 25N/mm2), DIN 1045.

The ground has to be of sufficient load capacity and free of abnormally high vibrating stresses. Keep the ground of installation free of external interfering sources. In case of outside vibrations, construction of a vibration-isolated foundation is recommended.

Check the hall ground for proper dimensioning and re-inforcement according to the specified load values. Maximum bend of the hall ground amounts 0.005 mm/m with dynamic load. The total bend, however, must not exceed 0.05 mm related to the total length and total width.

The user has to guarantee these values.

If necessary, changes to meet existing conditions at the site of installation are in order as long as static conditions are not affected.

No special foundation is necessary for the coolant system, hydraulic package and electrical cabinets provided the existing floor is even and of adequate load capacity.

Static and dynamic load specifications of the foundation

Centre of gavity: S1- Even load distribution on 8 levelling jacks

* Component: Machine base
* Load force: Maximum 95 kN
* Load: Static load

Centre of gravity: S2:load distribution on 4 levelling jacks

* Component: Rotary table
* Load force: Maximum 35 kN + weight of the workpiece
* Load: Static load

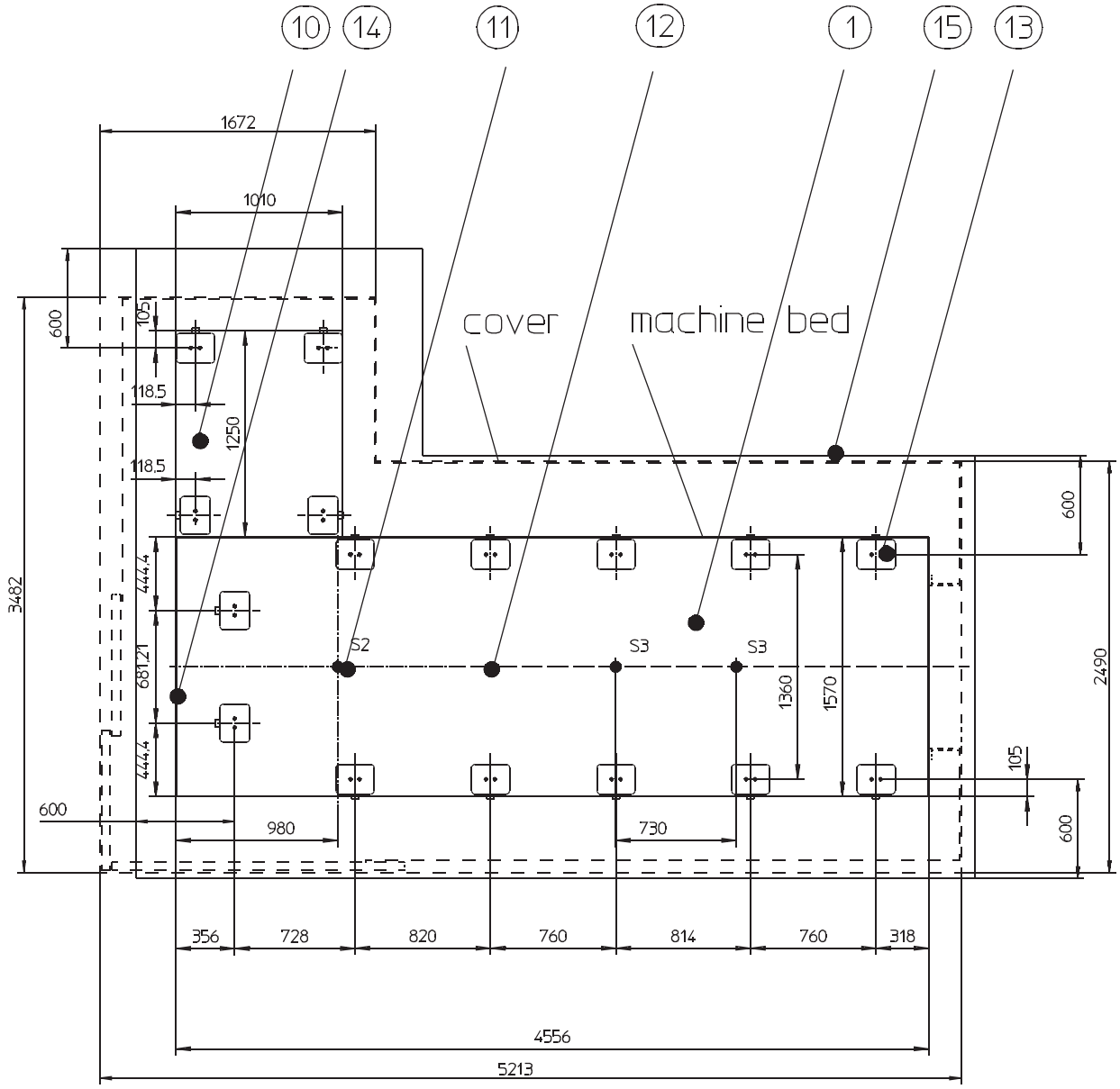
Centre of gravity: S3

* Component: Column
* Load force: Maximum 85 kN
* Load: Dynamic load

Centre of gravity: S4 - load distribution on 4 levelling jacks

* Component: Outer column bed
* Load force: Maximum 40 kN
* Load: Static load

Total static load: 255 kN Netto.



Dims in mm

Legend:
- 1: Machine base
- 10: Outer column bed
- 11: Positions of the weight gravity points of the grinding machine
- 12: Centre line for the machine base
- 13: Positions and dimensions of the levelling jacks (included in the delivery)
- 14: Front edge of the machine base
- 15: Concrete plate

Illustration 13: Foundation diagram for the grinding machine